Brief B: Can we 'stop' climate change?

1 Executive Summary

In accordance with the GICCE's commitment to "improving public understanding of Earth's climate and environment", our team has been producing a simple mathematical model linking the relationships between economic activity, greenhouse gas emissions, and their impact on the environment. This report will include an introduction describing the current context and three target hypothetical scenarios, four sections for mathematical modelling and data analysis, discussion section for limitations of the models, feasibility of achieving each target and summary of environmental impacts and finally a references list for citing all the sources used.

Our modelling show that to limit warming to 4°C by 2100 requires an annual carbon intensity reduction of around 2.5%. 2°C by 2050 requires a reduction of 3.0% to 3.5%, and the 1.5°C target by 2100 demands an unprecedented 13% annual reduction for the rest of the century. Without action, cumulative emissions will exceed even the 4°C budget, driving severe long-term impacts. Projected sea level rise ranges from 621mm under 4°C to 157mm under 1.5°C, while ocean pH could fall to 7.89 in the worst case, threatening coral reefs and key marine species.

2 Introduction

2.1 Background

Climate change refers to the long term alteration of Earth's climate system, primarily driven by anthropogenic greenhouse gas emissions from activities such as fossil fuel combustion, industrial processes and land conversion. The accumulation of carbon dioxide (CO_2) in the atmosphere has been the dominant forcing agent, trapping outgoing infrared radiation and increasing global mean surface temperatures. Since the pre-industrial era, this warming has already reached approximately 1.1°C contributing to significant and measurable impacts on the cryosphere, hydrosphere and biosphere.

A critical feature of climate change is its cumulative nature: total historical emissions determining the extent of future warming, with higher concentrations of atmospheric CO_2 leading to amplified effects on the climate system. Among the most significant consequences are sea level rise, resulting from thermal expansion and ice mass loss and ocean acidification, caused by the uptake of CO_2 by seawater. These processes threaten biodiversity, destabilise ecosystems and impose significant economic and social costs, underscoring the urgency of coordinated global mitigation strategies.

We have been given three scenarios to analyse as conditions for "stopping" climate change. These are:

- Global mean temperature rises to 4°C above pre-industrial levels and plateaus at that level from 2100.
- Global mean temperature rises to 2°C above pre-industrial levels and plateaus at that level from 2050.
- Global mean temperature rises no further than 1.5°C above pre-industrial levels this century and other aspects of climate change such as sea level rise stop by 2100.

The most ambitious target of 1.5°C is the target set out by the Paris Agreement and ratified by almost all UN signatory countries, but is commonly analysed by papers such as *Focus on cumulative emissions, global carbon budgets and the implications for climate mitigation targets* [1] to be unlikely to be achieved with current global de-carbonisation efforts.

With consideration of the serious environmental consequences awaiting inaction, this paper focuses on projecting impacts on sea levels and ocean acidification under each scenario, providing quantitative predictions to guide policy development and targeted mitigation strategies.

2.2 Key Relationships

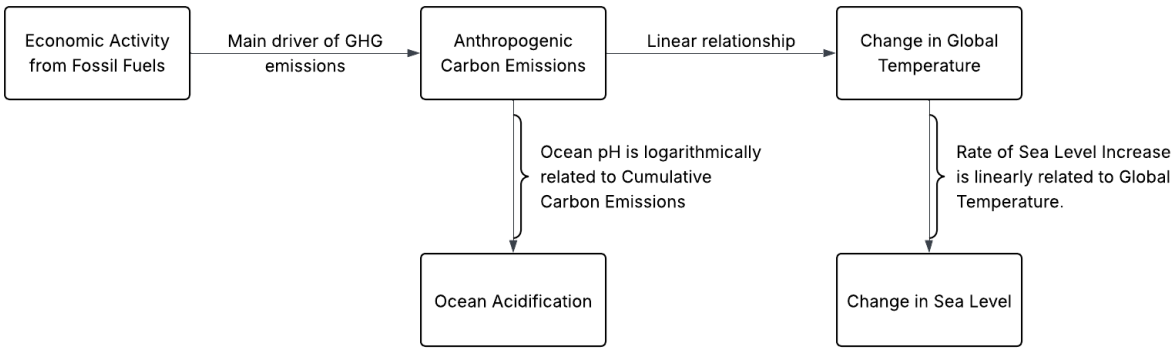

Figure 1: A Chart summarising the interactions of the key variables

The relationship between economic activity and anthropogenic greenhouse gas emission is the most central, and also the most variable.

3 Overview of Carbon Climate Response

3.1 Mathematical Modelling

CCR is a simplified metric that estimates how global temperature increases in response to cumulative carbon dioxide (CO_2) emissions.

CCR is defined as

$$\text{CCR} = \frac{\Delta T}{\Delta C_A} \times \frac{\Delta C_A}{E_T} = \frac{\Delta T}{E_T},$$

where:

- ΔT is the change in average global temperature (in °C) attributable to carbon,
- ΔC_A is the change in atmospheric carbon, in trillion tonnes of carbon (TtC),
- E_T is cumulative carbon in the form of CO_2 emissions, also in TtC.

The units of the metric are °C per TtC, i.e. change in degrees per trillion tonnes of carbon emitted.

3.2 Datasets

To independently reproduce the relationship proposed in the paper, we mainly use the *NASA* Global Temperature Anomaly Dataset[2] and the *Our World in Data* CO_2 Emissions Dataset[3]. These datasets overlap between the years 1880 and 2023, and all values were aligned to this common time range.

3.3 Usage in overall model

Note that CCR was introduced primarily as a method of establishing the linear relationship between cumulative carbon emissions and global temperature anomaly attributable to carbon, and hence that adapting it for use in estimating emissions targets (i.e. extending the model for non-carbon sources of heating) introduces a large amount of uncertainty. Thus an extension requires the assumptions that greenhouse gas composition and relative forcing efficiencies remain roughly constant, and that there are no other significant factors to changes in global temperature.

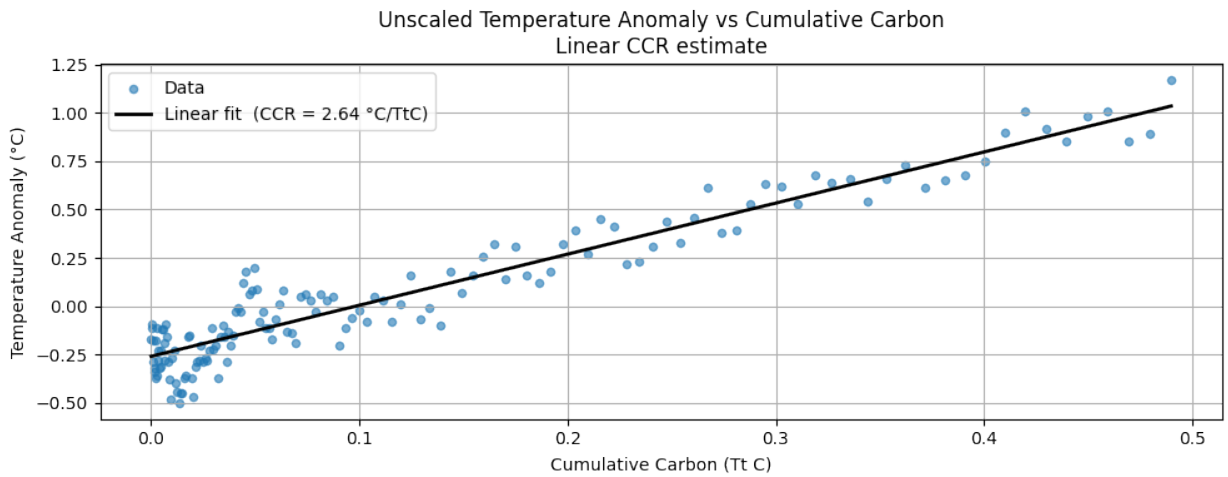

Figure 2: "CCR" of global temperature anomaly vs cumulative carbon emissions (unscaled anomaly).

Thus for the purposes of extrapolating cumulative carbon emissions to future total temperature anomaly, we have an effective "CCR" of 2.64°C TtC^{-1}. This is then used to approximate the remaining carbon budget by plugging temperature targets into

$$\text{Carbon Budget (TtC)} = \frac{\text{Target Temp (°C) - Current Temp (1°C)}}{\text{CCR}}$$

Resulting in the follow carbon budgets:

- At the targeted temperature of 4.0°C, the remaining budget is 1.14TtC out of a total 1.52TtC.
- At the targeted temperature of 2.0°C, the remaining budget is 0.38TtC out of a total 0.76TtC.
- At the targeted temperature of 1.5°C, the remaining budget is 0.19TtC out of a total 0.56TtC.

4 Modelling Emissions

4.1 Reference Model: Kaya's Identity

Decoupling economic growth and carbon emissions[4] by John Deutch explores Kaya's Identity originally proposed by energy economist Yoichi Kaya, as a framework for analysing the sources of CO_2 emissions. The identity decomposes carbon dioxide emissions into four components, is given by

$$\text{CO}_2 = P \times \frac{\text{GDP}}{P} \times \frac{E}{\text{GDP}} \times \frac{\text{CO}_2}{E}$$

where:

- P: Population (people)
- $\frac{\text{GDP}}{P}$: GDP per capita (USD/person)
- $\frac{E}{\text{GDP}}$: Energy intensity (energy per USD)
- $\frac{\text{CO}_2}{E}$: Carbon intensity (tonnes CO_2/unit energy).

4.2 Datasets

This report draws on datasets from *Our World in Data* (OWID) and the *World Bank*[5] to parameterise the Kaya's Identity model. CO_2 emissions data, sourced from OWID's *CO_2 and Greenhouse Gas Emissions*[3] (based on the Global Carbon Project), covers fossil fuel combustion and industrial activity. Population figures were obtained from OWID's demographic datasets[6], while energy data came from OWID's *Global*

Energy Substitution series[7], expressed in substituted energy equivalents (TWh). GDP data was retrieved from the World Bank and adjusted for inflation and PPP. These harmonised datasets enable consistent historical decomposition and scenario modelling.

These datasets collectively enabled the calculation of annualised percentage changes in the Kaya's Identity factors such as population, GDP per capita, energy intensity, and carbon intensity, forming the empirical basis for the model simulations.

4.3 Analysis Tool: LMDI Decomposition

The Logarithmic Mean Divisia Index(LMDI)[8] is a decomposition method used to analyse changes in aggregate indicators, such as carbon dioxide emissions, by breaking them down into contribution from multiple driving factors. LMDI is widely applied in energy and climate studies due to its perfect decomposition property (no residual terms), path independence and ability to handle zero values without distorting results

$$\Delta C = \Delta C_P + \Delta C_{GDP/P} + \Delta C_{E/GDP} + \Delta C_{C/E}.$$

Applying the LMDI onto each factor x in Kaya's Identity yields

$$\Delta C_x = \sum_{t=1}^{T} \frac{C_t - C_{t-1}}{\ln(C_t) - \ln(C_{t-1})} \cdot \ln\left(\frac{x_t}{x_{t-1}}\right).$$

where C_t denotes total CO_2 emissions at time t and x_t is the value of the selected factor in year t.

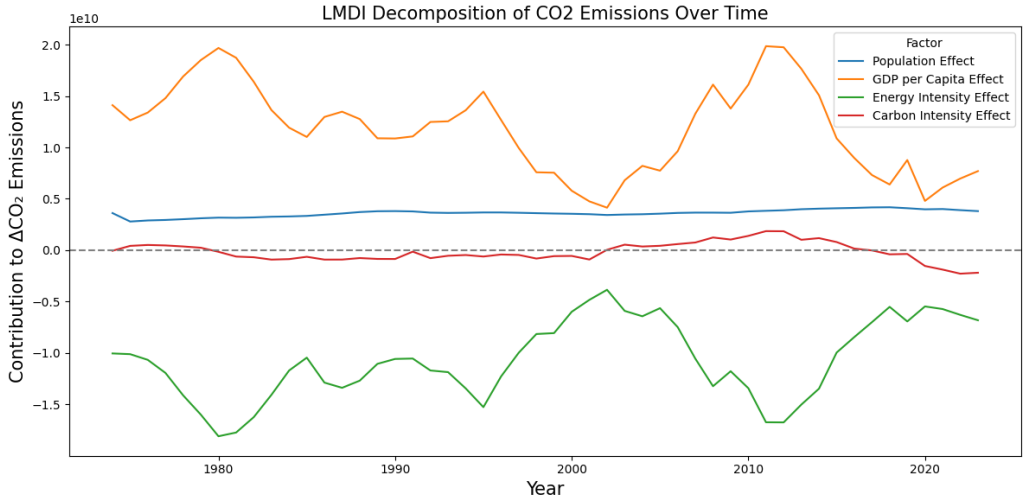

Figure 3: LMDI decomposition of global CO_2 emissions (1970–2023). GDP per capita is the largest driver of emissions growth, population has a stable positive contribution and energy intensity improvements have partially offset this trend.

This methodology yields key quantitative insights into CO_2 emissions:

- **GDP per capita effect** is consistently positive, making it the largest driver of emissions growth across the entire period. Short term fluctuations align with global economic cycles, but the long term trend reflects the sustained economic expansion
- **Population effect** is steadily positive with little variation, reflecting persistent global population growth over the period
- **Carbon intensity effect** remains close to zero for much of the time series, indicating minimal historical impact on emissions. This suggests that large scale decarbonisation of energy sources has yet to significantly influence global totals

- **Energy intensity effect** is consistently negative, indicating that improvements in energy efficiency have acted as a persistent offset to emissions growth. While the magnitude varies over time, the downward contribution is one of the most stable mitigating factors observed.

While it is theoretically possible to reduce emissions by limiting GDP growth or population growth, doing so would hinder global development and is therefore not a sustainable approach. Efforts to improve energy efficiency have already delivered significant reductions in emissions, demonstrating their effectiveness. This leaves carbon intensity as the primary variable that can be further influenced. Strengthening efforts to reduce carbon intensity, by transitioning to low carbon energy sources, will have a decisive impact on the trajectory of future emissions and is central to addressing climate change.

4.4 Predictive Model and Simulation

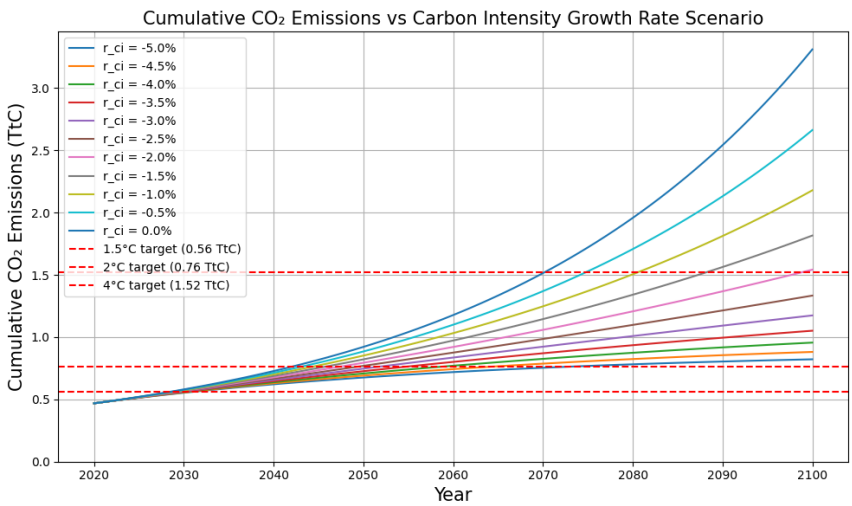

Figure 4: Cumulative CO_2 emissions from 2020 to 2100 under different carbon intensity growth rates r_{ci} ranging from -5% to 0% per year, with all other factors in Kaya's Identity consistent with historical data. Horizontal dashed lines indicate the cumulative carbon budgets associated with 1.5 °C (0.56 TtC), 2 °C (0.76 TtC), and 4 °C (1.52 TtC) global warming targets.

Moving forward with the insights from the LMDI decomposition, the emissions trajectory can be modelled exponentially from the sum of the annualised percentage changes in the contributions of the Kaya's Identity components, expressed as

$$C(t) = C_0 \cdot (1 + r)^t$$

where:

- $C(t)$: Annual Emission at t years
- C_0: Emission at year $t = 0$ or in this case 2020
- $r = r_{population} + r_{\text{GDP per capita}} + r_{\text{energy intensity}} + r_{\text{carbon intensity}}$.

r	ΔCO_2
$r_{\text{population}}$	+1.68%
$r_{\text{GDP per capita}}$	+5.89%
$r_{\text{energy intensity}}$	-4.79%

Table 1: r values obtained from LMDI decomposition on historical data.

By fixing $r_{\text{population}}$, $r_{\text{GDP per capita}}$, and $r_{\text{energy intensity}}$ to their historical averages, the only variable left to adjust in scenario modelling is $r_{\text{carbon intensity}}$. This approach isolates the role of decarbonisation in shaping future cumulative emissions pathways.

4.5 Results from Model

The model was simulated under a range of annual carbon intensity change rates (r_{ci}) from -5% (rapid decarbonisation) to 0% (no change), isolating the effect of decarbonisation efforts on future cumulative emissions trajectories.

Scenario	**Target date**	**Model-derived r_{ci}**
1.5 °C limit	2100	-13.0%
2 °C plateau	2050	-3.5% to -3.0%
4 °C plateau	2100	-2.5%

Table 2: Model-derived carbon intensity change rate (r_{ci}) requirements to meet each warming scenario under the constant-rate assumptions of the model.

It should be noted that the -13.0% annual reduction requirement for the 1.5 °C limit was not included in Figure 4 but was derived using the same model framework and assumptions as the other scenarios.

5 Overview of Sea Level Rise and Temperature Anomalies

5.1 Mathematical Modeling

The 2007 paper by Rahmstorf [9]provides a model where the rate of rise of global sea levels and increase in global surface temperature is proportional, given by the differential equation

$$\frac{dH}{dt} = a(T - T_0),$$

Where H is the global mean sea level in millimetres, t is time in years and a is the proportionality constant in millimetres per year per degrees Celsius, T is the global mean temperature in degrees Celsius and T_0 is the previous equilibrium temperature value in degrees Celsius. This equation calculates the change of the mean sea level above the previous equilibrium state. The researchers noted a very high level of agreement between the computed value for global sea levels using the equation and the actual observed sea levels, with differences always below 1 cm.

5.2 Data Gathered

The data for both sea level rise[10] and sea surface temperature[11] are taken from OurWorldInData.org. Two graphs are shown below, one compares the rise in sea level relative to the average sea levels between 1993 and 2008 and sea surface temperature anomalies, relative to average sea surface temperatures between 1961 and 1990. The second graph is a linear fit for the scatter plot of global sea level rise vs sea surface temperature increases over the same year.

The data for sea level rise[12] is taken as an average between the researchers Church and White and a study by the University of Hawaii Sea Level Center. Data are collected using tide gauges and satellite altimeters. Tide gauges work by measuring the high tide and low tides from around the world and calculating an average.

Satellite altimeters send a radar pulse at the ocean and measure the speed and intensity of the pulse returned.

The data for sea surface temperature[13] is taken from the Met Office Hadley Centre. They collect the data through a combination of buoy and ship measurements, collected globally.

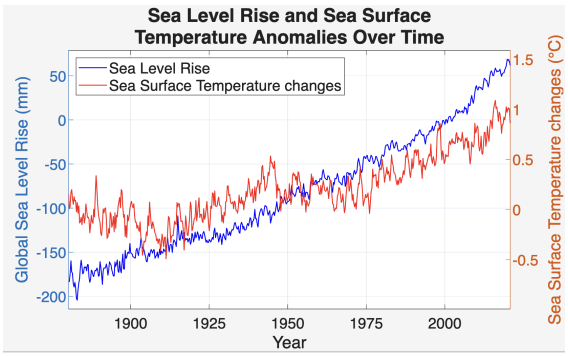

Figure 5: Global Sea Level Rise and Global sea surafce temperature increase over time, with a shared temporal x-axis for easy visual comparison

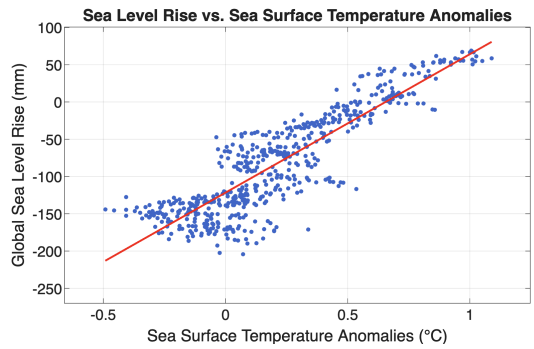

Figure 6: Global sea level rise vs Sea Surface Temperature Anomalies from 1850-2025, given as both a scatterplot and linear fit regression line

Then using Coefficient of Determination to analysis its accuracy, R^2

$$R^2 = 1 - \frac{SSE}{SST}.$$

- SSE is a error function which reflects variation of data about the line of best fit.
- SST is the difference between the data with the average data.

The scatterplot has a coefficient of determination value of

$$R^2 = 0.78,$$

indicating a strong fit between the data points and trend line. The least squares regression line is

$$y = 185.73x - 121.60.$$

This indicates that the proportionality constant a described earlier would be 185.73, showing that global sea levels rise increases by roughly 185.73 millimeters per year per degree Celsius.

The data shows that the model is relatively accurate, since the linear trend line between global sea level rise and sea surface temperature increases fits strongly, with R^2 value of 0.78. This means that roughly 78% of the variance in the rise of sea levels can be explained by the change in sea surface temperature anomalies.

6 Overview of Greenhouse Gas Emissions and Ocean Acidification

6.1 Mathematical model

Ocean acidification is a direct consequence of rising atmospheric CO_2 levels, driven largely by anthropogenic greenhouse gas emissions. Caldeira (2003) [14] simulated historical and projected ocean chemistry from 1975 to 2300 using the Lawrence Livermore National Laboratory ocean general-circulation model, finding a clear inverse relationship between seawater pH and atmospheric CO_2 concentrations. The underlying chemistry is well established: when CO_2 dissolves in seawater, it forms carbonic acid, which dissociates to release hydrogen ions H^+, lowering pH (Zeebe, 2001 [15])

$$CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-.$$

From equillibrium chemistry, the relationship between hydrogen ion concentration and aqueous CO_2 is given by

$$K = \frac{[HCO_3^-][H^+]}{CO_2(aq)}.$$

Since K is constant, $H^+ \propto CO_2$ and because pH is defined as $pH = -\log[H]^+$, the theoretical relationship between pH and atmospheric CO_2 can be expressed as

$$pH = m - n \log CO_2.$$

where m is the initial pH and n is the constant of proportionality.

6.2 Datasets

Atmospheric CO_2 concentration data were obtained from "Greenhouse gas emissions" [3], while ocean surface pH data were sourced from "Physical and bio-geochemical modulation of ocean acidification in the central North Pacific" [16]. For consistency, the dataset was restricted to the period 1988–present.

A logarithmic model was fitted to relate atmospheric CO_2 concentration to ocean surface pH, yielding:

$$pH = 10.11 - 0.34 \log CO_2(concentration).$$

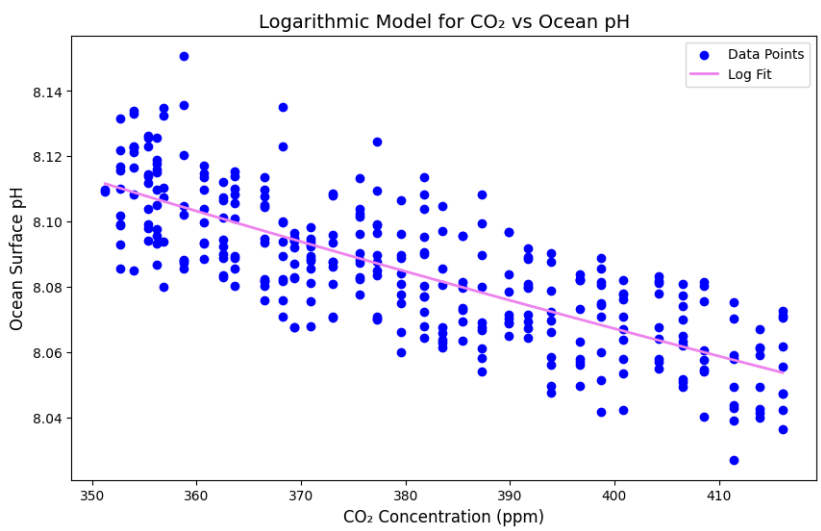

Figure 7: Logarithmic fit showing the inverse relationship between atmospheric CO_2 concentration and ocean surface pH. Higher CO_2 levels correspond to lower pH values, indicating progressive ocean acidification.

This relationship was then linked to CO_2 emission via

$$CO_2(concentration) = 255.18 \cdot CO_2(emission) + 297.56$$

$$pH = 10.11 - 0.34 \log(255.18 \cdot CO_2(emission) + 297.56).$$

The model achieved an R^2 value of 0.594, indicating that approximately 59.4% of the variation in ocean surface pH can be explained by atmospheric CO_2 concentrations. This strong inverse correlation aligns with theoretical expectation from carbonate chemistry that rising CO_2 levels will progressively lower seawater pH, with significant implications for marine ecosystems.

7 Discussion

7.1 Feasibility

Using the carbon intensity reduction rates (r_{ci}) derived from the predictive modelling framework, the following discussion interprets these model outputs in terms of their real-world decarbonisation requirements and likelihood of attainment. Most importantly, the predictive model demonstrates that inaction ($r_{\text{ci}} = 0.0\%$) leads to a continued exponential rise in cumulative emissions, ultimately exceeding the carbon budget for even the most lenient target scenario and locking in severe long-term climate impacts.

Scenario 1 (4°C plateau by 2100). Requiring a carbon intensity reduction rate of -2.5% annually, this pathway represents a moderate-to-high global decarbonisation challenge. It exceeds historical performance but remains technically achievable with sustained global policy efforts, large-scale renewable deployment, and steady energy efficiency gains.

Scenario 2 (2°C plateau by 2050). With a required r_{ci} between -3.5% and -3.0% annually, this pathway demands high global decarbonisation rates. Achieving it would require rapid infrastructure turnover, accelerated phase-out of carbon-intensive technologies, and significant coordination between governments, industries, and markets.

Scenario 3 (1.5°C this century; other impacts cease by 2100). This pathway's required r_{ci} of -13.0% annually is unprecedented and approaches infeasibility under current technological, political, and economic constraints. It would require transformative breakthroughs, immediate large-scale carbon removal deployment, and systemic changes in global energy consumption and production patterns.

7.2 Limitations

7.2.1 Carbon Climate Response

CCR, as applied here, compresses the emissions–temperature link to a single constant slope, $\text{CCR} = \Delta T / E_T$. This treats the airborne fraction $\Delta C_A / E_T$ and the carbon-attributable share of temperature anomaly as fixed, so changes in sinks or other forcings are folded into error. Because CCR depends only on cumulative totals, it ignores emissions timing and ocean heat uptake, so different pathways with the same E_T can yield different ΔT. Our fitted value (2.64°C TtC^{-1}) is sensitive to the period, baseline and whether ΔT is scaled to carbon-attributable warming.

7.2.2 Emissions Modelling

This model builds sequentially on Kaya's Identity, LMDI decomposition and exponential projection, offering a transparent and reproducible framework for linking emission outcomes to specific drivers. By fixing population growth, GDP per capita growth, and energy intensity improvements to historical averages, the analysis isolates carbon intensity r_{ci} as the primary policy lever, highlighting decarbonisation as the central factor in maintaining a safe emissions trajectory. The approach is interpretable, adaptable and builds off of well established methods, however it assumes long term stability in the fixed drivers and constant reduction rates, potentially overlooking disruptions, non-linear dynamics and policy or market changes. While it does not capture the full complexity of the global system, the model effectively underscores the central challenge: sustaining a decay based emissions pathway rather than an exponentially compounding problem.

7.2.3 Sea Level Rise and Sea Surface Temperature anomalies

Despite the linear model having good levels of agreement with past data, the researchers in Rahmstorf 's 2007 paper admit that the model becomes less accurate with future data. The model is expected to overestimate

the sea levels in 2100 by around 120mm. The researchers state that the model might not be robust for future data, due to uncertain factors in the future, such as disappearing glaciers that will cause less contribution to rising sea levels, or the ice sheets might respond more strongly to increasing temperatures in the 21st century than in the 20th century.

7.2.4 Greenhouse Gas Emission and Ocean Acidification

The analysis supports the predicted inverse relationship between ocean pH and atmospheric CO_2.The model does not account for the significant modulation of pH changes caused by buffering by carbonate and biological processes. The data-driven models, like Gaussian Process Regression, developed by Dehghani et al. (2024) [17] had $R^2 > 0.98$, which was higher than the $R^2 = 0.549$ found in this study. These machine learning models outperformed analytical methods in capturing complex nonlinear interactions, including temperature, pressure, CO_2 concentration, and solution volume. This suggests that incorporating such techniques could improve the realism and accuracy of future ocean acidification research.

7.3 Impacts

This section analyses the impact on the environment with regards to sea level changes and ocean acidification, according to the three scenarios given in the introduction.

- Scenario 1: global mean temperature rises to 4°C above pre-industrial levels and plateaus at that level from 2100.
- Scenario 2: global mean temperature rises to 2°C above pre-industrial levels and plateaus at that level from 2050.
- Scenario 3: global mean temperature rises no further than 1.5°C above pre-industrial levels this century and other aspects of climate change such as sea level rise stop by 2100.

7.3.1 Sea Level Rise

	Sea Level Rise Relative to Average Between 1993 and 2008 (mm)	Cities at risk of flooding	Glacier loss
Scenario 1	621.32	Miami, Jakarta and Bangkok	90%
Scenario 2	249.86	Shanghai, New Orleans and Mumbai	70%
Scenario 3	157.00	Tokyo, London and New York City	50%

Table 3: A table summarising the overall sea level rise for the scenarios and impacts on glaciers and cities

7.3.2 Ocean Acidification

	pH	Species at Risk	Coral Reef Losses
Scenario 1	7.89	Calcified algae, Corals, Molluscs, and Echinoderms	30%–35%
Scenario 2	8.00	Calcified algae, Corals, Molluscs	15%-25%
Scenario 3	8.04	Molluscs, Corals	10%-15%

Table 4: A table summarising the ecological impacts of various species at different pH levels

References

[1] H. D. Matthews, K. Zickfeld, R. Knutti, and M. R. Allen. Focus on cumulative emissions, global carbon budgets and the implications for climate mitigation targets. *Environmental Research Letters*, 2018.

[2] NASA. Global surface temperature. https://climate.nasa.gov/vital-signs/global-temperature/?intent=121, 2025. [Accessed 05 June 2025].

[3] Hannah Ritchie. Co2 emissions dataset. https://ourworldindata.org/co2-dataset-sources, 2022. [Accessed 05 June 2025].

[4] John Deutch. Decoupling economic growth and carbon emissions. *Joule*, 1(1):3–5, 2017.

[5] World Bank. Gdp per capita (current us$). https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?end=2024&start=1960&view=chart, 2025.

[6] Global Change Data Lab. Population, 2024. Based on UN World Population Prospects (2024); processed by Our World in Data. https://ourworldindata.org/grapher/population.

[7] Global Change Data Lab. Global primary energy consumption by source (substitution method), 2025. Data from Energy Institute – Statistical Review of World Energy (2025), processed by Our World in Data. https://ourworldindata.org/grapher/global-energy-substitution.

[8] B.W. Ang. The lmdi approach to decomposition analysis: a practical guide. *Energy Policy*, 33(7):867–871, 2005.

[9] Stefan Rahmstorf. A semi-empirical approach to projecting future sea-level rise. *Science*, 315(5810):368–370, 2007. [Accessed 15 June 2025].

[10] Our World in Data. Climate change: Sea level rise. https://ourworldindata.org/explorers/climate-change?time=1914-09-15..latest&Metric=Sea+level+rise&Long-run+series=false&country=ATA~Gulkana+Glacier~Lemon+Creek+Glacier~OWID_NAM~South+Cascade+Glacier~Wolverine+Glacier~Hawaii~Arctic+Ocean~OWID_WRL, 2025. [Accessed 15 June 2025].

[11] Our World in Data. Climate change: Sea surface temperature anomaly. https://ourworldindata.org/explorers/climate-change?time=1914-09-15..latest&Metric=Sea+surface+temperature+anomaly&Long-run+series=false&country=ATA~Gulkana+Glacier~Lemon+Creek+Glacier~OWID_NAM~South+Cascade+Glacier~Wolverine+Glacier~Hawaii~Arctic+Ocean~OWID_WRL, 2025. [Accessed 15 June 2025].

[12] National Oceanic and Atmospheric Administration (NOAA). Climate change: Global sea level, 2023. [Accessed 15 June 2025].

[13] Colin P. Morice, John J. Kennedy, Nick A. Rayner, J. P. Winn, Emma Hogan, R. E. Killick, R. J. H. Dunn, Timothy Osborn, Philip Jones, and I. R. Simpson. An updated assessment of near-surface temperature change from 1850: the hadcrut5 dataset. *Journal of Geophysical Research: Atmospheres*, 126(3), 2019. [Accessed 16 June 2025].

[14] Ken Caldeira and Michael E Wickett. Anthropogenic carbon and ocean ph. *Nature*, 425(6956):365–365, 2003.

[15] Richard E Zeebe and Dieter Wolf-Gladrow. *CO2 in seawater: equilibrium, kinetics, isotopes*, volume 65. Gulf Professional Publishing, 2001.

[16] Hannah Ritchie, Pablo Rosado, and Veronika Samborska. Climate change impacts data explorer – ocean acidification: mean seawater ph. https://ourworldindata.org/explorers/climate-change?Metric=Ocean+pH&Long-run+series=false&country=..., 2025. Accessed: 2025-08-09.

[17] Mohammad Rasool Dehghani, Moein Kafi, Hamed Nikravesh, Maryam Aghel, Erfan Mohammadian, Yousef Kazemzadeh, and Reza Azin. Data driven models for predicting ph of co2 in aqueous solutions: Implications for co2 sequestration. *Results in Engineering*, 24:102889, 2024.

8 Appendix

8.1 Sea Level Rise and Sea Surface Temperature Anomalies

Code from Matlab used:

Global Sea Level Rise and Global sea surface temperature increase over time graph:

```
% Load the datasets
seaLevelData = readtable('sea-level-rise.csv');
seaTempData = readtable('world_sea_surface_temperature.csv');

% Convert 'Day' columns to datetime for both datasets
seaLevelData.Day = datetime(seaLevelData.Day, 'InputFormat', 'yyyy-MM-dd');
seaTempData.Day = datetime(seaTempData.Day, 'InputFormat', 'yyyy-MM-dd');

% Merge the datasets on the 'Day' column
[~, idxLevel, idxTemp] = intersect(seaLevelData.Day, seaTempData.Day);
mergedData = table(seaLevelData.Day(idxLevel), ...
                seaLevelData.("GlobalSeaLevelAsAnAverageOfChurchAndWhite_2011_AndUHSLCData")(idxLevel), ...
                seaTempData.("MonthlySeaSurfaceTemperatureAnomalies")(idxTemp), ...
                'VariableNames', {'Day', 'SeaLevel', 'SeaTemp'});

% Create a figure with two y-axes
figure;
yyaxis left;
plot(mergedData.Day, mergedData.SeaLevel, 'b-', 'LineWidth', 1);
ylabel('Global Sea Level Rise (mm)', 'FontSize', 16);
ylim([min(mergedData.SeaLevel) - 10, max(mergedData.SeaLevel) + 10]);
set(gca, 'FontSize', 20);

yyaxis right;
plot(mergedData.Day, mergedData.SeaTemp, 'r-', 'LineWidth', 1);
ylabel('Sea Surface Temperature changes ( C )', 'FontSize', 16);
ylim([min(mergedData.SeaTemp) - 0.5, max(mergedData.SeaTemp) + 0.5]);
set(gca, 'FontSize', 16);

% Add title and legend
title({'Sea Level Rise and Sea Surface', 'Temperature Anomalies Over Time'}, 'FontSize', 20);
legend('Sea Level Rise', 'Sea Surface Temperature changes', ...
       'Location', 'northwest', 'FontSize', 16);
grid on;

% Adjust the x-axis
xlim([min(mergedData.Day), max(mergedData.Day)]);
datetick('x', 'yyyy', 'keeplimits');
xlabel('Year', 'FontSize', 18);
```

Listing 1: MATLAB code for sea level and temperature analysis

Global Sea Level Rise vs Global sea surface temperature increase over time linear fit graph:

```
%%
% Load the datasets
seaLevelData = readtable('sea-level-rise.csv');
seaTempData = readtable('world_sea_surface_temperature.csv');

% Convert 'Day' columns to datetime for both datasets
seaLevelData.Day = datetime(seaLevelData.Day, 'InputFormat', 'yyyy-MM-dd');
seaTempData.Day = datetime(seaTempData.Day, 'InputFormat', 'yyyy-MM-dd');

% Merge the datasets on the 'Day' column
[~, idxLevel, idxTemp] = intersect(seaLevelData.Day, seaTempData.Day);
mergedData = table(seaLevelData.Day(idxLevel), ...
                seaLevelData.("GlobalSeaLevelAsAnAverageOfChurchAndWhite_2011_AndUHSLCData")(idxLevel), ...
```

```
                seaTempData.("MonthlySeaSurfaceTemperatureAnomalies")(idxTemp), ...
                'VariableNames', {'Day', 'SeaLevel', 'SeaTemp'});

% Create a scatter plot with SeaTemp on x-axis and SeaLevel on y-axis
figure;
scatter(mergedData.SeaTemp, mergedData.SeaLevel, 20, 'filled', ...
        'MarkerFaceColor', [0.2 0.4 0.8], 'MarkerEdgeColor', 'none');
xlabel('Sea Surface Temperature Anomalies ( C )', 'FontSize', 20);
ylabel('Global Sea Level Rise (mm)', 'FontSize', 20);
title('Sea Level Rise vs. Sea Surface Temperature Anomalies', 'FontSize', 24);
grid on;

% Add linear fit to show correlation
hold on;
p = polyfit(mergedData.SeaTemp, mergedData.SeaLevel, 1);
f = polyval(p, mergedData.SeaTemp);
plot(mergedData.SeaTemp, f, 'r-', 'LineWidth', 2);

% Calculate R  (coefficient of determination)
SS_res = sum((mergedData.SeaLevel - f).^2);
SS_tot = sum((mergedData.SeaLevel - mean(mergedData.SeaLevel)).^2);
R2 = 1 - (SS_res / SS_tot);

% Calculate correlation coefficient (R)
R = corrcoef(mergedData.SeaTemp, mergedData.SeaLevel);
R_value = R(1,2);

% Set font size for axes ticks (larger)
set(gca, 'FontSize', 16, 'Box', 'on');

% Adjust figure margins for better display
set(gcf, 'Color', 'w');
```

Listing 2: MATLAB code for sea level vs temperature linear fit graph